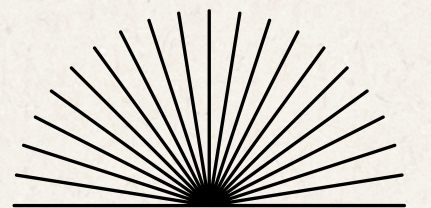




CREDIT RISK PROJECT

by Lin Zhu Emory University

PRESENTED ON:
11/17/2025



Agenda

01	Overview
02	Data Cleaning
03	Data Analysis
04	Feature Engineering
05	Model Selection & Tuning
06	Model Evaluation



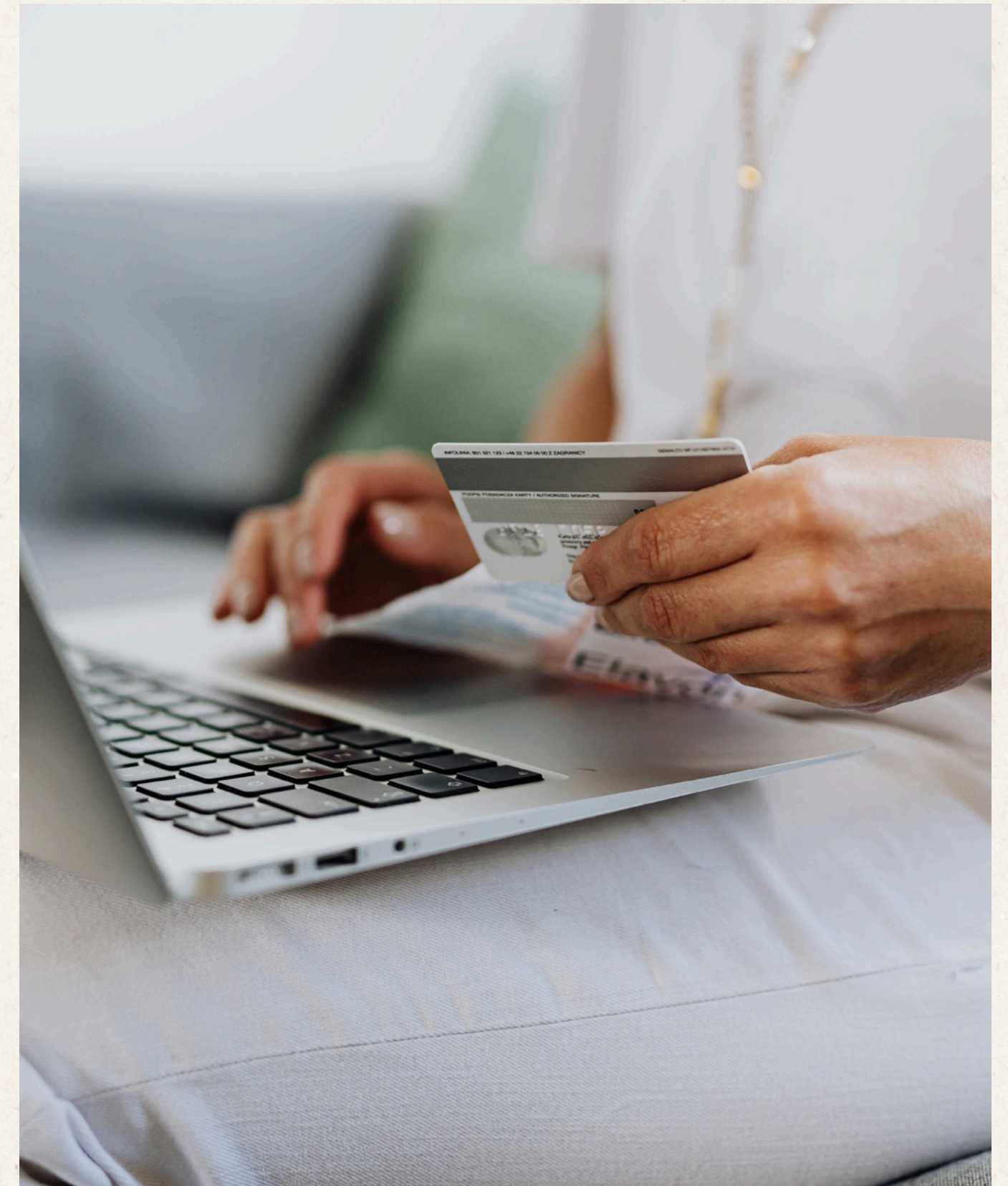
Overview

This project builds a fraud-risk prediction model for a consumer credit-loan company. Each year, the firm receives hundreds of thousands of applications, and even a small share of fraudulent users can cause large financial losses. Traditional rule-based checks capture only limited fraud and do not scale.

By combining applicant demographics with historical credit-behavior patterns, our goal is to develop a data-driven model that identifies high-risk applicants early, reduces potential loss, and supports faster, more accurate credit-approval decisions.

01 Dataset Overview

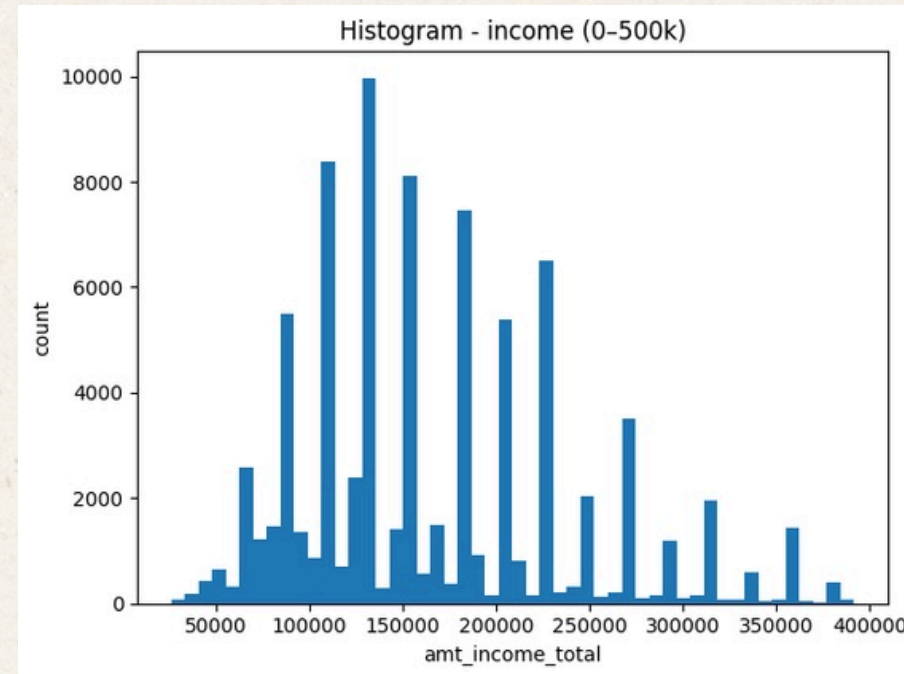
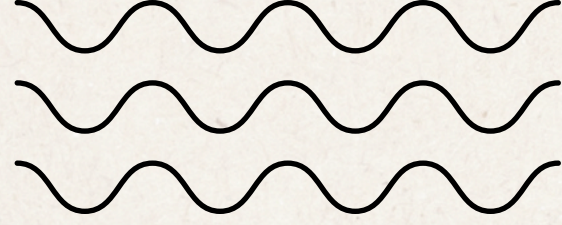
- **Two data sources**
 - Application Record: 406,230 applicants with 18 demographic & financial fields
 - Credit Record: 1,048,575 monthly credit status entries (1-to-many with applicants)
- **Merged dataset:**
 - 85,089 applicants with both application and credit history
 - Contains aggregated behavioral features derived from credit-status sequences
- **Feature types:**
 - Numerical: income, children count, family size, phone/email flags
 - Categorical: gender, education, employment type, family status, housing type
 - Time-series credit behavior: months_balance, repayment status codes
- **Key challenge:**
 - Significant class imbalance (fraud cases are rare but costly)
 - Many applicants have incomplete or inconsistent credit-record coverage





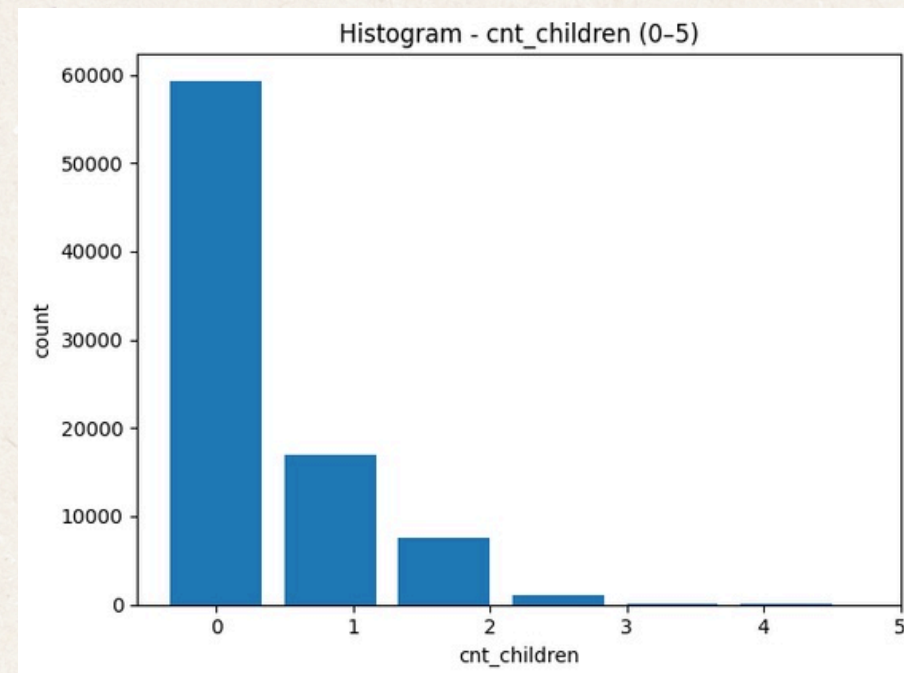
02 **Data Cleaning**

- Removed duplicate applicants based on unique ID + email
- Handled missing values via median imputation & selective column removal
- Standardized formats:
 - Converted timestamps → datetime
 - Normalized categorical strings (lowercase, trimming)
 - Ensured consistent numeric types
- Processed outliers in income & financial indicators using winsorization
- Merged credit-record sequences into user-level aggregated features
- Addressed class imbalance as preparation for modeling (SMOTE / reweighting later)



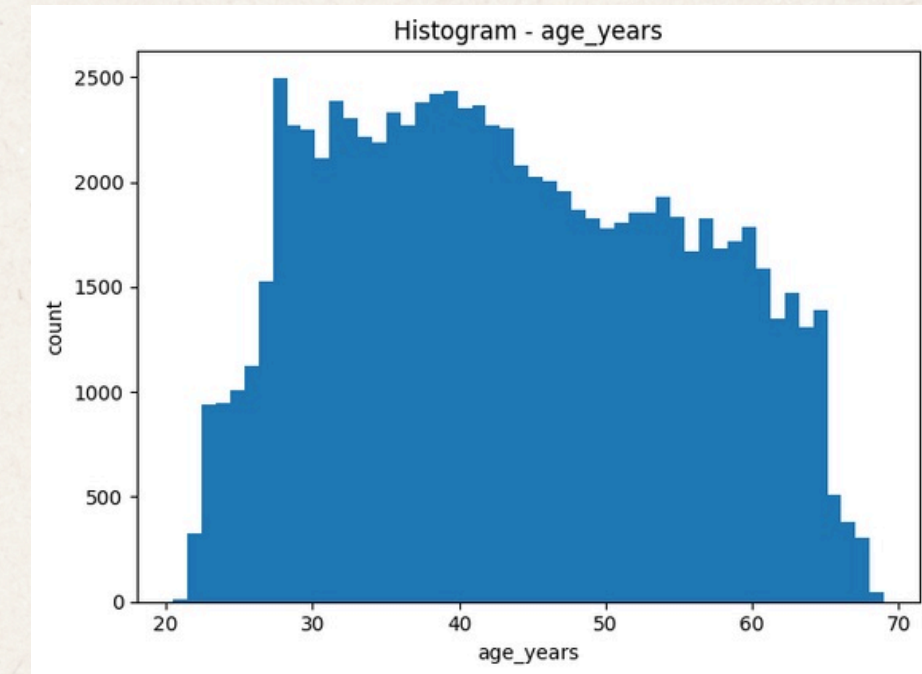
Income

- Histogram zoomed to 0–500k; most applicants earn around 72k–220k.
- 2,596 records (3.1%) lie above the IQR upper fence (>393,750), forming a long right tail with a few values up to 6.75M.



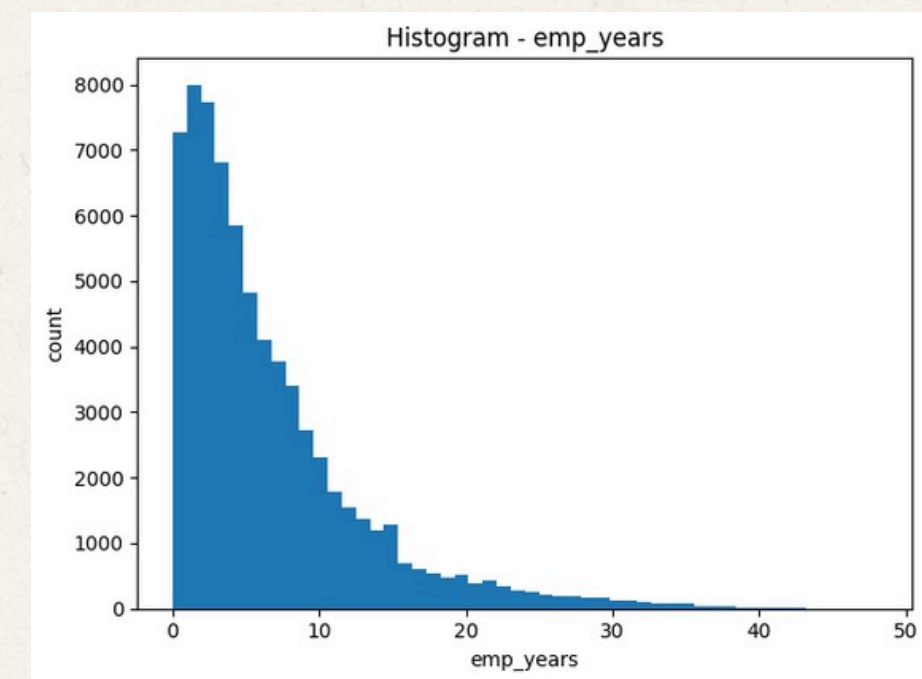
Children

- Plot zoomed to 0–5 children, covering 99.99% of applicants.
- Most applicants have 0–2 children (median = 0); only 8 records (0.009%) exceed 5 children, a very rare extreme group.



Age

- Mean around 44, mainly 33–53 years old.
- Typical working-age population, with few very young or very old applicants.



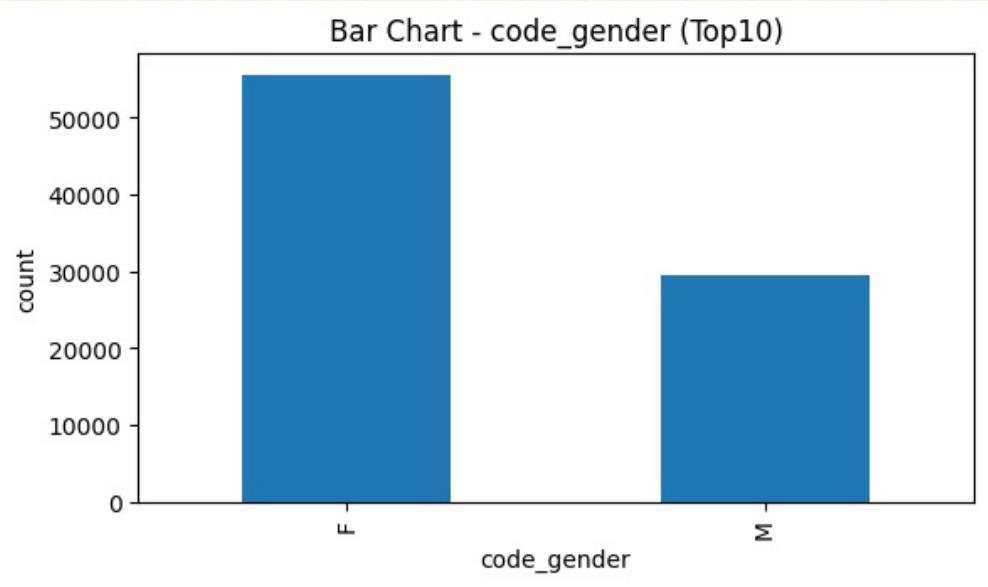
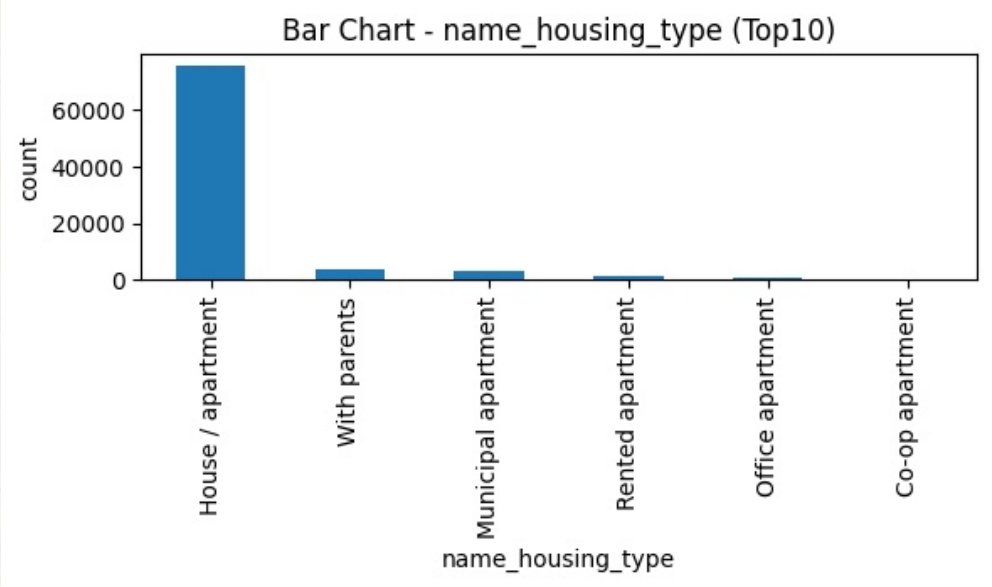
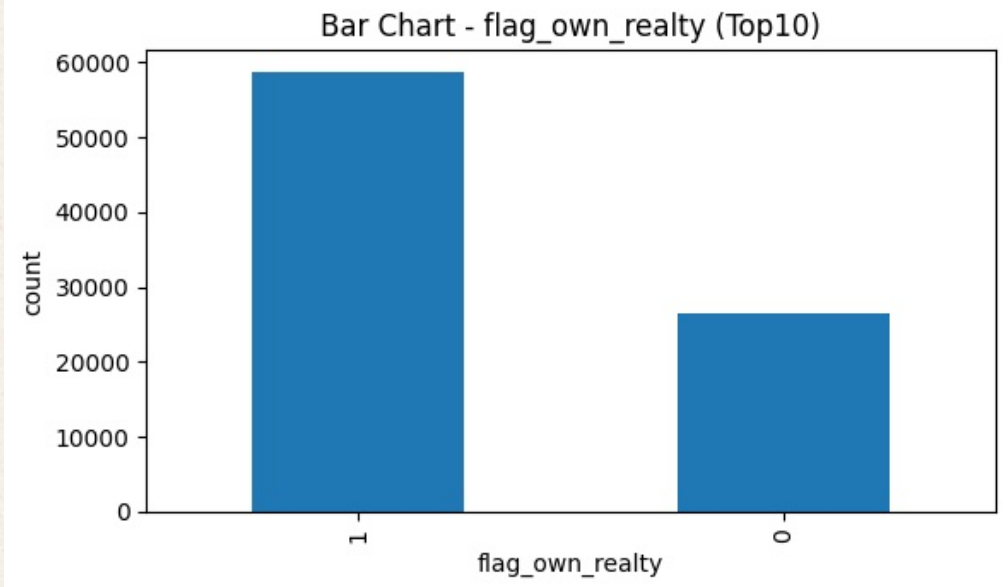
Employment

- emp_years is mostly within 0–10 years, with a long right tail.
- System code days_employed = 365243 is used for retired people and is not included in the hisogram.

04

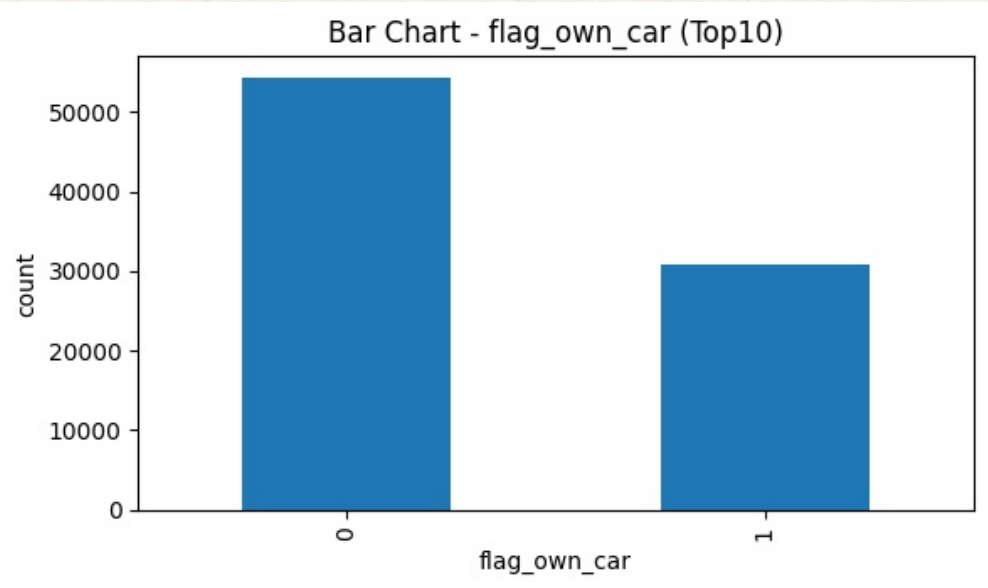
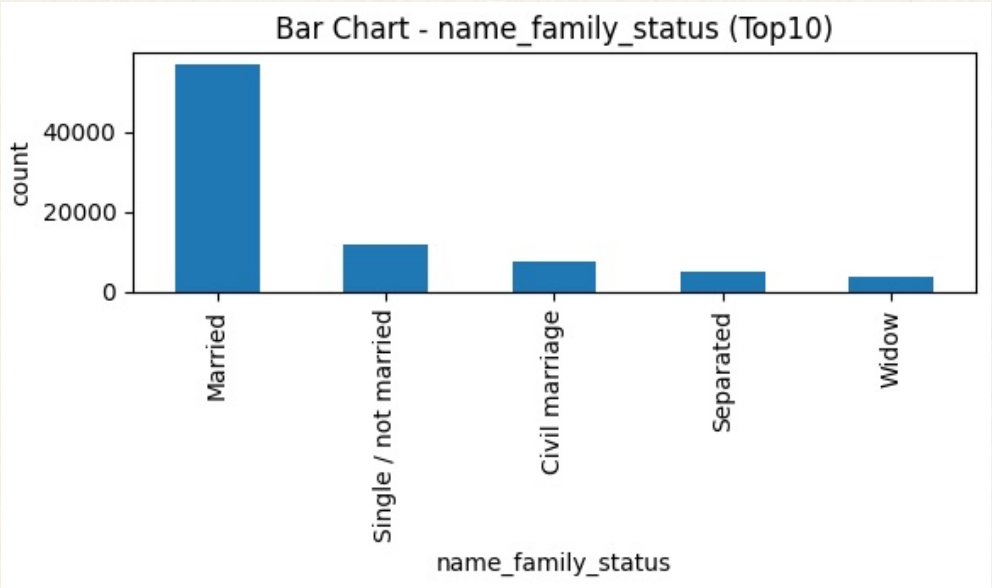
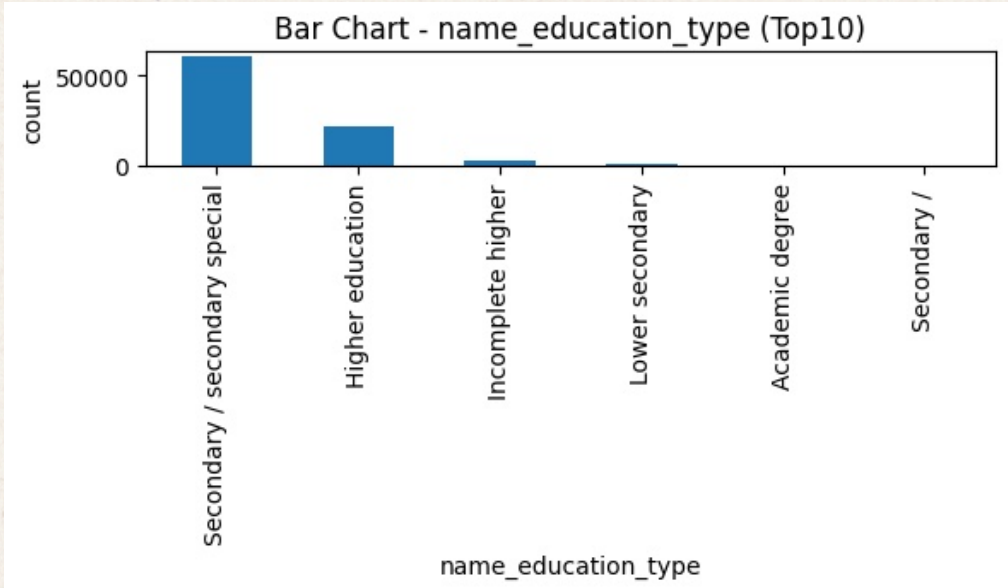
Data Analysis

Categorical Features: Applicant Profile



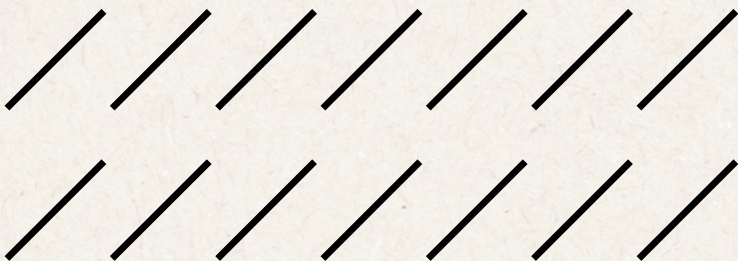
Demographics & housing

- Majority of applicants are female.
- Most applicants own at least one property.
- Housing is dominated by house / apartment; living with parents or in rented / municipal / office apartments is rare.



Education, family & car ownership

- Education is highly skewed: about 70% secondary and 25% higher education.
- Around two-thirds of applicants are married; other statuses are small groups.
- Car ownership is lower – more applicants do not own a car than do.



1. Sample profile

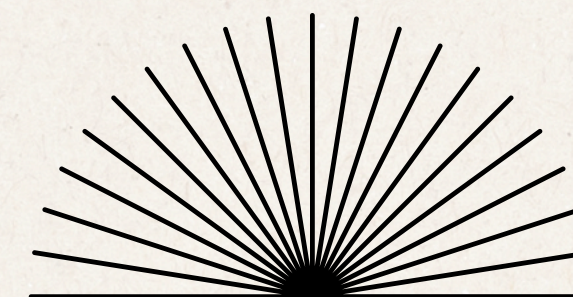
- Applicants are mostly mid-40s, married, with secondary education and at least one property.
- This looks more like a “stable, mainstream” borrower segment than a broad population sample.

2. Distribution & imbalance

- Key numeric variables (income, children, employment years) are heavily right-skewed with visible tails.
- Several categorical variables have dominant majority classes and large “Other” buckets → strong class imbalance.

3. Data quality signals

- Special codes such as `days_employed = 365243` encode “no current employment” instead of true tenure.
- Rare extreme patterns (e.g., many children, very high income) exist but represent only a tiny fraction of the data.



Feature Engineering



Account Length

How long has the bank actually known this customer?

- Longer account history = more stable past behavior.
- Shorter history = limited information, higher uncertainty.

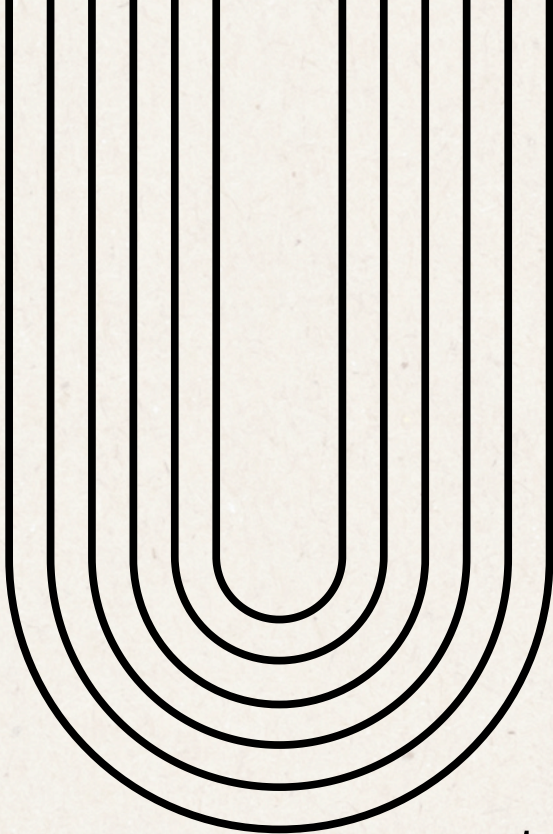
Unemployment Indicator

Captures whether the person has stable income



Age & Employment

- **Age** (age_years)
Converted from days_birth.
- **Employment Length** (years_employed)
Converted from days_employed; 365243 replaced as missing.
- **Employment Length Category** (employment_length_cat)
Bucketed employment years into <1yr, 1–5yrs, 5–10yrs, etc>
- **Work–Age Ratio** (work_age_ratio & work_age_ratio_filled)
Measures how much of someone’s life has been spent working.
Strong stability indicator.
- **Filled Versions** (years_employed_filled, work_age_ratio_filled)
Missing values replaced safely using “Unemployed/Unknown”



Socioeconomic Features

- **IncomeEmployed**
1 = stable income source (working, commercial associate, state servant)
0 = pensioner or student.
- **EduHigherPlus**
1 = higher education or academic degree.
- **HousingOwned**
1 = owns property (house / co-op)
0 = rented / with parents / municipal housing.

We created a set of meaningful, interpretable, risk-related features that are cleaner and more predictive than raw data.



**Behavior / Credit
History Features**



**Age & Employment
Features**



**Socioeconomic
Features**

- account_length

Age & Employment Features

- Base Features
- age_years
- years_employed

Employment Stability Indicators

- unemployed
- has_employment_record

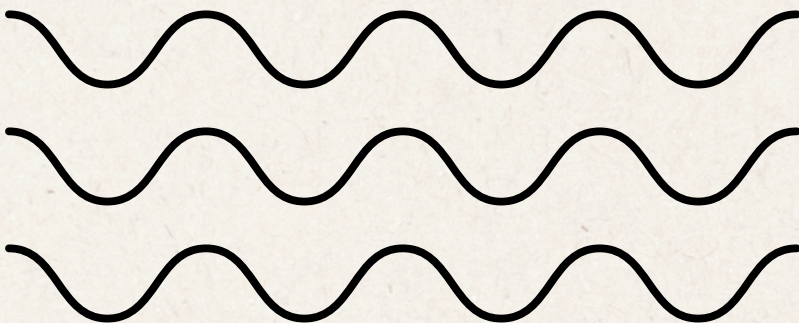
Derived Features (New)

- employment_length_cat
- work_age_ratio
- is_retirement_age

Filled / Processed Versions

- years_employed_filled
- work_age_ratio_filled

- IncomeEmployed
- EduHigherPlus
- HousingOwned



**Feature Set
Overview**

Identify the key audiences for which
the project is intended.

From Features To Models

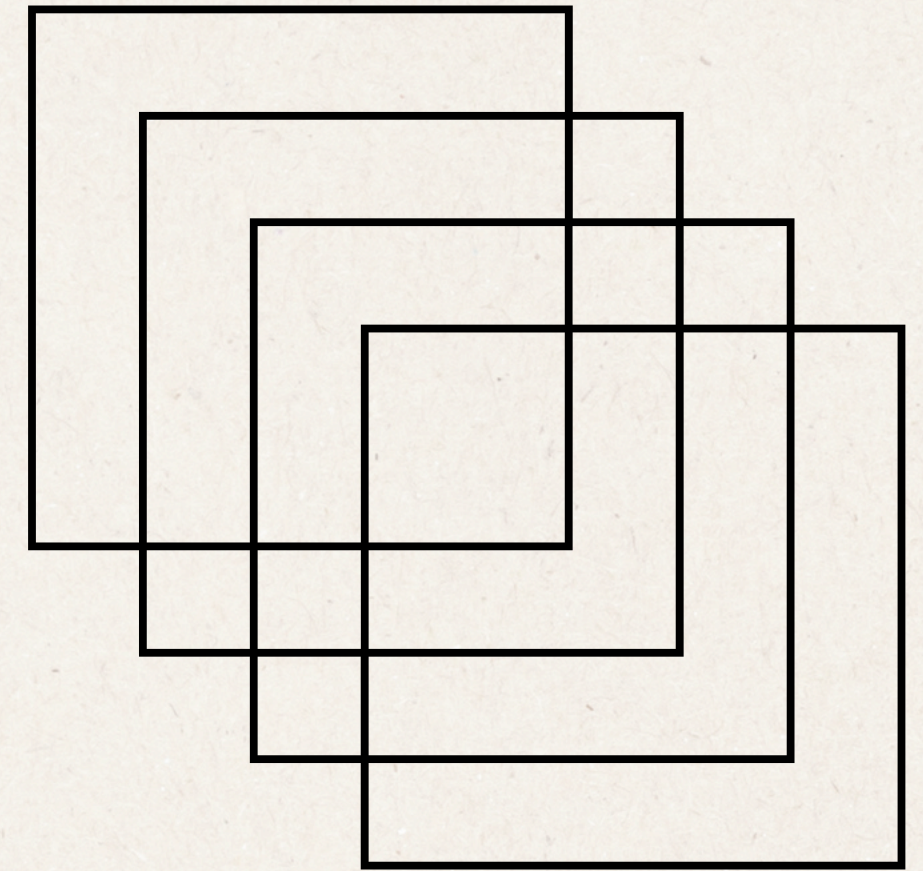
Preprocessing & Next Steps

Preprocessing

- **Fill missing values**
- **Encode Categories**
- **Standardize numerics**
- **Feature check**

Next Step?

- **Model choosing**
- **train and test split**
- **Model evaluation**
- **Feature decisions**



Modeling Strategy

Random Forest

Purpose: Baseline

Parallel ensemble with strong generalization, establishes performance benchmark

XGBoost

Purpose: Improvement

Gradient boosting with regularization, two rounds for tuning optimal recall

LightGBM

Purpose: Improvement

Leaf-wise growth, best recall-precision balance with efficiency

Key Focus:

Maximizing recall for credit default detection while maintaining reasonable precision

Data Preparation & Hyperparameter Tuning

TRAIN/TEST SPLIT: 70/30 STRATIFIED SPLIT

SMOTE UPSAMPLING: BALANCED CLASS DISTRIBUTION

Model	Key Parameters	Best Configuration
Random Forest	max_depth, n_estimators	depth=10, trees=200, min_samples_leaf=16
XGBoost (Round 1)	learning_rate, n_estimators, max_depth	lr=0.1, trees=300, depth=5, subsample=1.0
XGBoost (Round 2)	Expanded search space	lr=0.13, trees=1500, depth=6, subsample=0.8
LightGBM	num_leaves, n_estimators, learning_rate	lr=0.05, trees=100, leaves=63, no depth limit

Model Performance Comparison

Model	Recall	Precision	F1-Score	AUC	Accuracy
Random Forest (Baseline)	0.8129	0.9598	0.8803	—	0.8894
XGBoost (Initial)	0.8509	0.989	0.9147	0.9427	0.9207
XGBoost (Tuned) ✓	0.8679	0.9535	0.9087	0.9435	0.9128
LightGBM	0.8438	0.9921	0.9119	0.9478	0.9185

Key Findings

XGBoost (Tuned) Selected as Final Model

Highest recall (0.8679) for maximum default detection with strong precision (0.9535) and AUC (0.9435)

Tuning Impact:

- Round 1: Recall 0.8509, AUC 0.9427
- Round 2: Recall 0.8679 (+1.7%), expanded n_estimators to 1500 and adjusted learning rate to 0.13

Why XGBoost Wins:

- Achieves highest recall for catching credit defaults while maintaining 95% precision
- Strong regularization prevents overfitting on imbalanced credit data

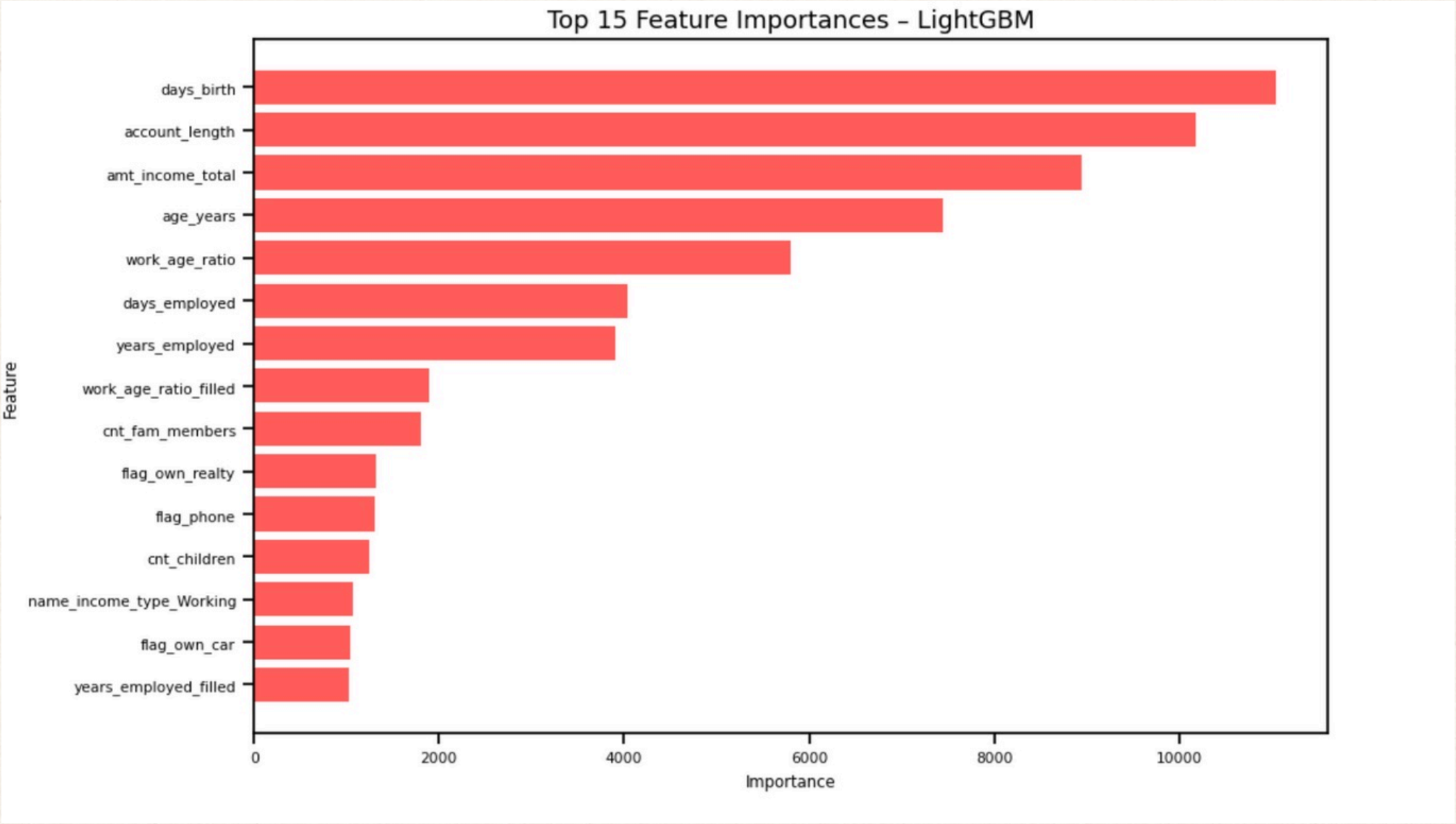
Bottom Line:

XGBoost with tuned hyperparameters achieves the best recall-precision balance for credit card default prediction

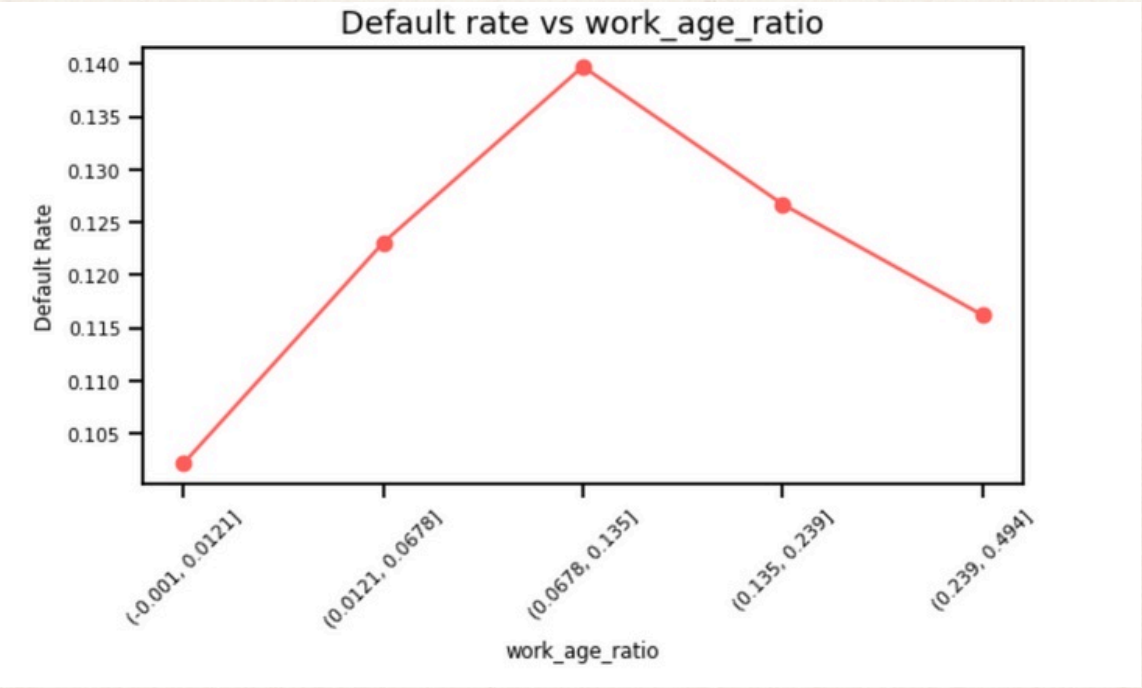
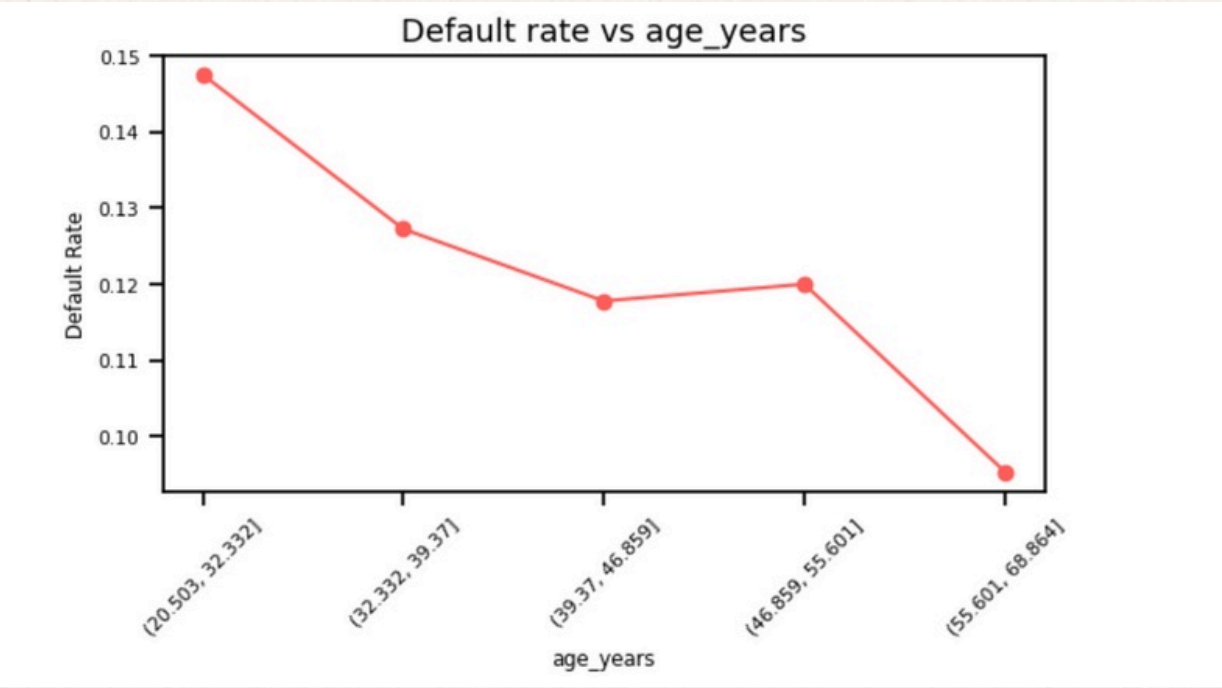
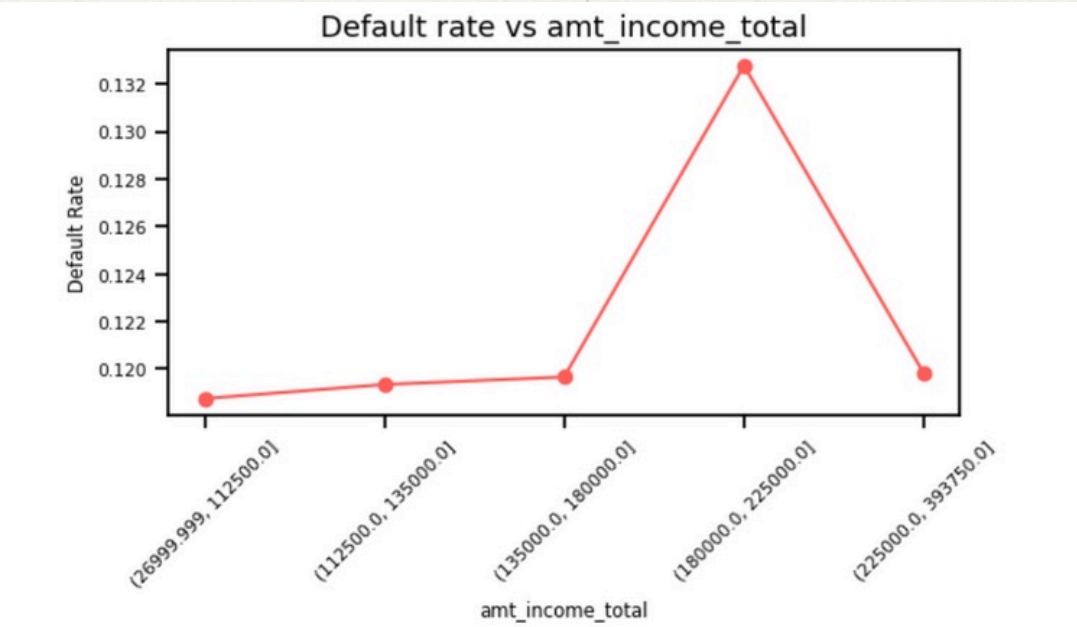
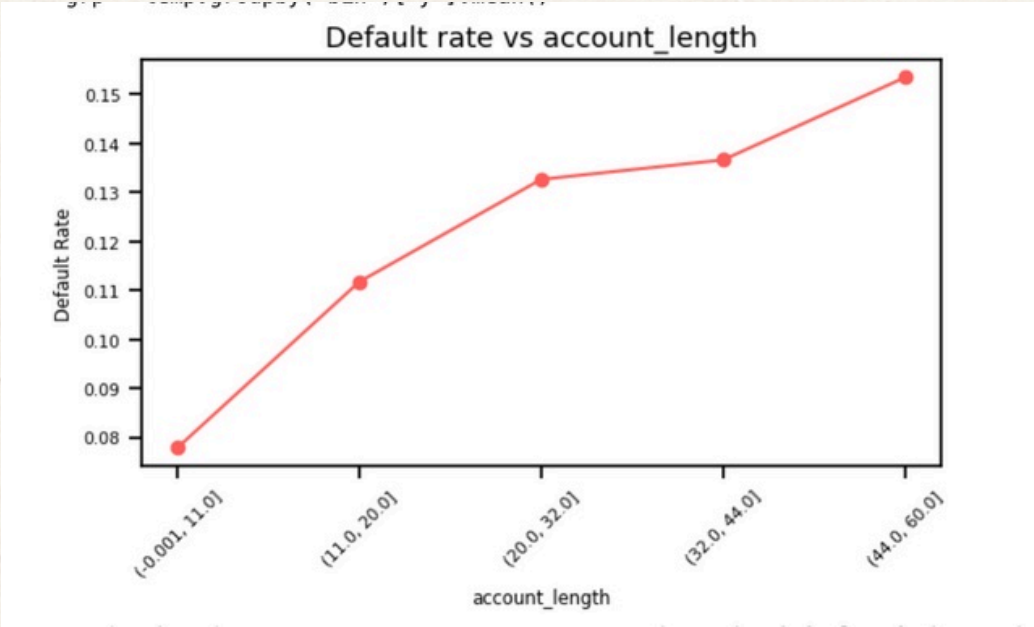
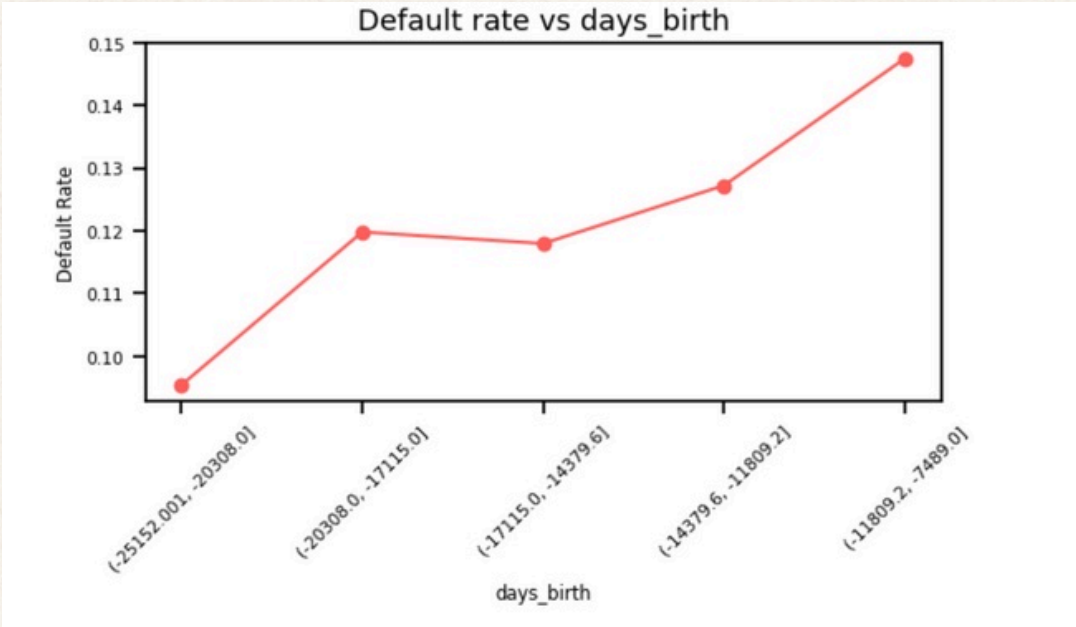
Model Evaluation

Metric	LightGBM
AUC	0.948
Precision	0.992
Recall	0.844
F1-score	0.912

- LightGBM was chosen as the final model because it captures more fraud cases (**high recall**)
- while minimizing false alarms (**very high precision**)
- making it the most reliable for real-world credit-risk screening.



Feature Importance



Commercial Value

Business Assumptions

(for a mid-sized online consumer-lending platform)

- Annual credit applications: 500,000
- Estimated fraud rate: 2%
→ 10,000 fraudulent applications / year
- Average exposure per successful fraud: \$20,000
- Current rule-based system recall: 50%
- New LightGBM model recall: **84.4%**

Fraud Detection – Before vs. After

- **Old rule-based system (Recall = 50%)**

$$\text{Frauds_Caught_Old} = 10,000 \times 0.50 = 5,000$$

- **New LightGBM model (Recall = 0.844)**

$$\text{Frauds_Caught_New} = 10,000 \times 0.844 = 8,440$$

- **Incremental fraud cases detected**

$$\text{Incremental} = 8,440 - 5,000 = 3,440$$



Monetary Impact

Annual loss avoided thanks to higher recall

$$\text{Savings} = 3,440 \times \$20,000 = \$68,800,000 \approx \mathbf{\$69M}$$

in fraud losses prevented per year

Thank you